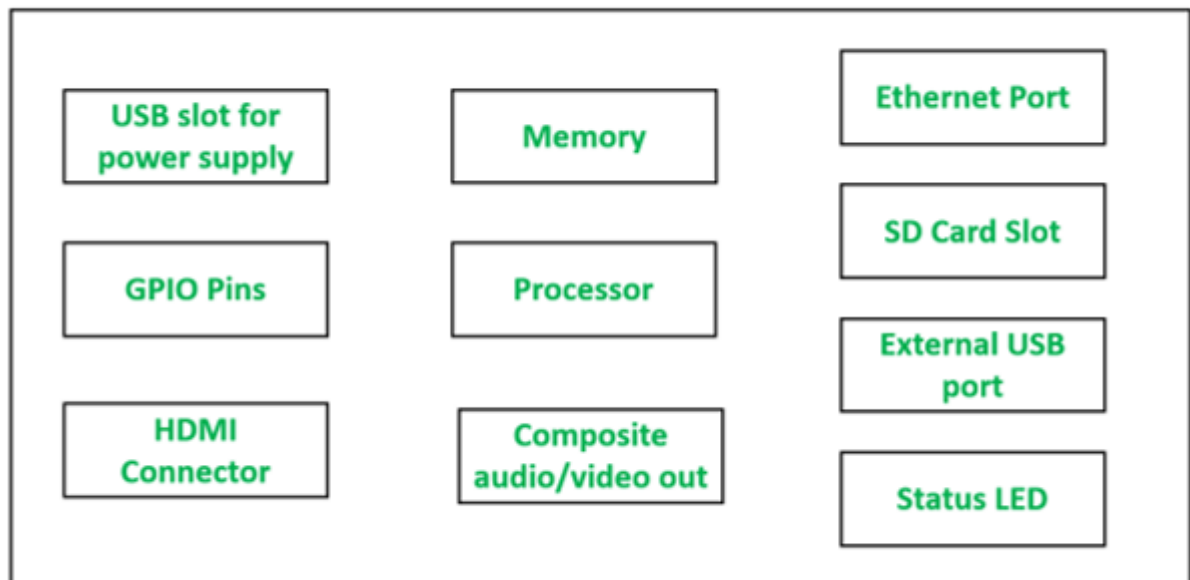


Single Board Computers -Raspberry-Pi

Raspberry Pi is developed by Raspberry Pi Foundation in the United Kingdom. The Raspberry Pi is a series of powerful, small single-board computers.

Raspberry Pi is launched in 2012 and there have been several iterations and variations released since then.

Various versions of Raspberry Pi have been out till date. All versions consist of a Broadcom system on a chip (SoC) with an integrated ARM-compatible CPU and on-chip graphics processing unit (GPU).

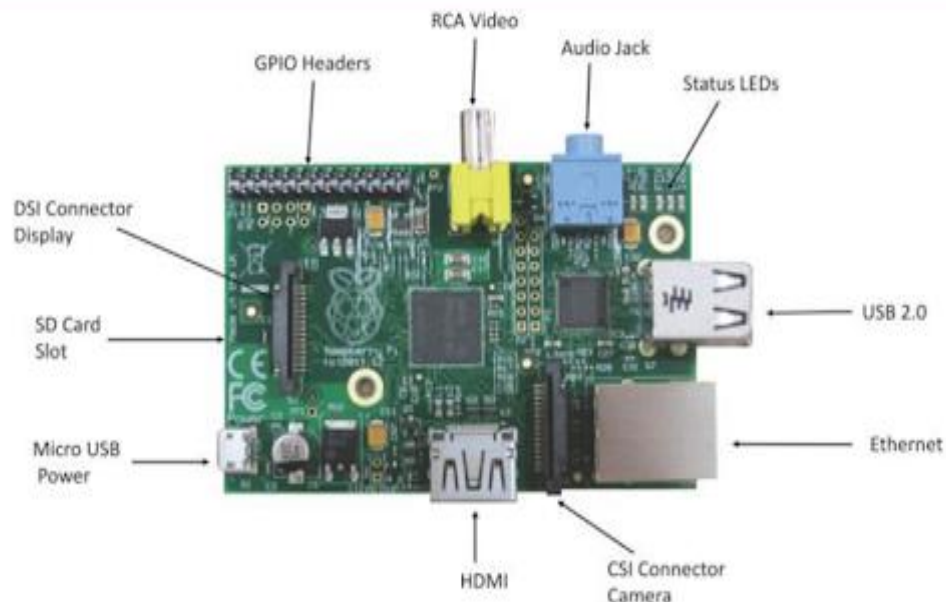


Block Diagram of Raspberry Pi

- A low-cost, **credit-card-sized** minicomputer.
- Plugs into a computer monitor or TV.
- Uses a standard keyboard and mouse.
- Use for explore computing as for programmings like Scratch and Python languages.
- Capable as a normal desktop computer to browsing the internet and playing high-definition video, to making spreadsheets, word-processing, and playing games.”

- Allowing interface sensors and actuators through the general purpose IO pins
- Supports Python out of box

Structure of board



Raspberry Pi board uses various components and peripherals as follows:-

Processor and Ram

- Based on an ARM processor.
- The latest version of Raspberry Pi (model B, revision 2) comes with a 700 MHz low-power ARM 1176JZ-F processor And a 512 MB SD Ram.

USB ports

- Two USB 2.0 USB ports on Raspberry pi can provide a current of up to 100 mA.
- For connecting devices that draw a current of more than 100 mA, an external USB-powered hub is required.

Ethernet port

- Standard RJ45 Ethernet port.
- Connect an ethernet cable or USB Wi-Fi adaptor to provide internet connectivity.

HDMI outputs

- The HDMI port on Raspberry Pi provides both video and audio output.
- Connect the Raspberry Pi to a monitor by an HDMI cable.
- For monitors that have a DVI port but no HDMI port so use an HDMI to DVI adaptor or cable.

Composite video output with RCA jack

- Support both PAL and NTSC video output.
- RCA jack is used to connect old television that has an RCA input only.

Audio output

- 3.5 mm audio output jack is used for audio output to old television along with the RCA Jack for video.
- The audio quality is inferior to the HDMI output.

Display serial interface(DSI)

- Used to connect an LCD panel to Raspberry Pi.

Camera serial interface(CSI)

- Used to connect a camera module to Raspberry Pi.

Status LEDs

- Raspberry Pi has 5 status LED is which are:-
- ACT:- SD card access.
- PWR:- 3.3 V power is present.
- FDX:- Full duplex LAN connected.
- LNK:- link/ network activity.
- 100:- 100 Mbit LAN connected.

SD card slot

- Not have a built-in operating system and storage so plug in an SD card loaded with a Linux image to an SD card slot.

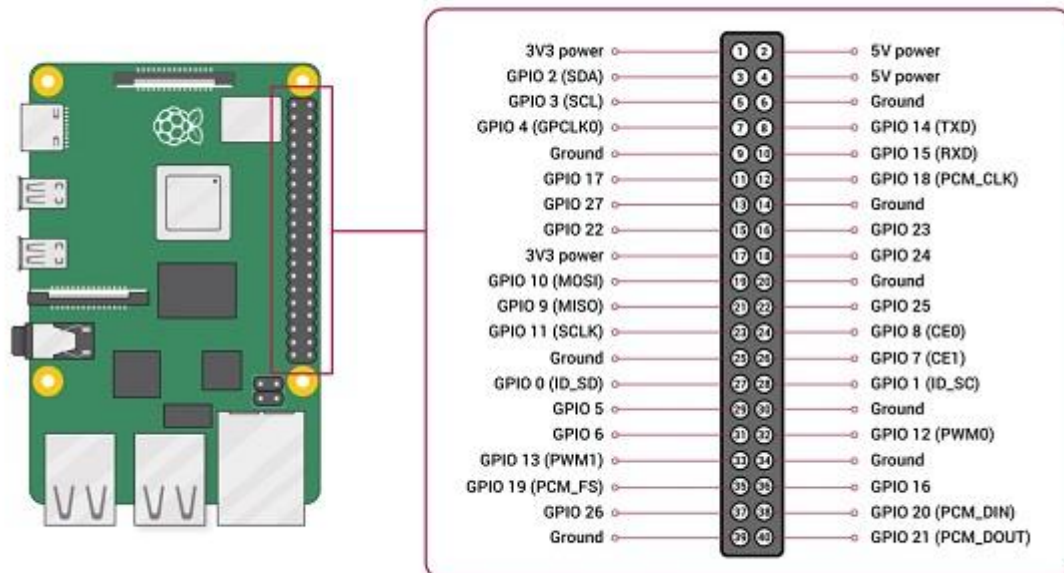
- Required at least an 8GB SD card for the setting up NOOBS software (new out-of-the-box software)

Power input

- A micro-USB connector for power input.

GPIO Pin

- A number of general-purpose Input / Output pins are used by Raspberry Pi.
- There are four types of pins on Raspberry Pi:- True GPIO pins, SPI interface pins, 12C interface pins, and Serial RX and TX pins.



Voltages

From the above diagram, we can see that there are two 5V pins and two 3V3 pins on the board. It also has several ground pins (0V). All these pins are unconfigurable.

Outputs

A GPIO pin can be designated as an output pin. The pin set as output pin can be set to 3V(high) or 0V(low).

Inputs

A GPIO pin can be designated as an input pin. The pin set as input pin can be read as 3V(high) or 0V(low). You can use internal pull-up or pull-down resistors.

above diagram, GPIO2 and GPIO3 pins have fixed pull-up resistors but for the other pins, you can configure it in software.

Alternative Functions

GPIO pins can be used with a variety of alternative functions. Among them, some are available on all pins and others on specific pins.

PWM: Pulse-width modulation

Software PWM are available on all the pins whereas Hardware PWM are available on GPIO12, GPIO13, GPIO18, and GPIO19.

SPI: Serial Peripheral Interface

The SPI are available on the following –

SPI0: MOSI (GPIO10); MISO (GPIO9); SCLK (GPIO11); CE0 (GPIO8), CE1 (GPIO7)

SPI1: MOSI (GPIO20); MISO (GPIO19); SCLK (GPIO21); CE0 (GPIO18); CE1 (GPIO17); CE2 (GPIO16)

I2C: Inter-integrated Circuit

The I2C are available on the following –

Data: (GPIO2); Clock (GPIO3)

EEPROM Data: (GPIO0); EEPROM Clock (GPIO1)

Serial

The serial function is available at the following –

TX(GPIO14)

RX(GPIO15)

Raspberry pi Connecting to the Cloud

Connecting a Raspberry Pi to the cloud allows you to access and manage your Raspberry Pi remotely, exchange data with cloud services, and perform various tasks such as data logging, remote control, and automation. Here's a general overview of how to connect a Raspberry Pi to the cloud:

1. **Select a Cloud Service Provider:** Choose a cloud service provider that fits your needs. Some popular options include AWS (Amazon Web Services), Microsoft Azure, Google Cloud Platform, and various IoT-focused platforms like AWS IoT, Azure IoT, or Google Cloud IoT. Each has its own set of features, pricing, and capabilities.
2. **Set Up Your Raspberry Pi:** Ensure your Raspberry Pi is set up correctly, connected to the internet, and running an appropriate operating system like Raspbian (now called Raspberry Pi OS).
3. **Install Necessary Software:** Depending on your chosen cloud platform, you may need to install specific libraries or SDKs on your Raspberry Pi. For example, if you're using AWS, you would install the AWS SDK for Python (Boto3). If you're using Azure, you might use the Azure IoT SDK or Azure Python SDK.
4. **Create Cloud Resources:** Create the necessary resources on your cloud platform. For example, if you're using AWS IoT, you'd create an IoT Thing, a policy, and certificates.
5. **Connect to the Cloud:** Write code on your Raspberry Pi to connect to your cloud platform of choice. This typically involves using the SDK you installed earlier and providing the necessary authentication credentials (e.g., API keys, certificates, or access tokens).
6. **Publish and Subscribe to Topics (MQTT):** Many IoT applications use the publish-subscribe model for communication. MQTT (Message Queuing Telemetry Transport) is a common protocol for this purpose. You can publish data from your Raspberry Pi to specific MQTT topics and subscribe to topics to receive commands or data from the cloud.
7. **Data Handling and Processing:** Once your Raspberry Pi is connected, you can send and receive data between the Raspberry Pi and the cloud. You may want to implement data processing, storage, or analytics depending on your project's requirements.
8. **Security Considerations:** Ensure proper security practices. Use secure connections (HTTPS, MQTT over TLS), secure your IoT devices and cloud resources, and regularly update your Raspberry Pi's software to patch any security vulnerabilities.

9. **Monitoring and Management:** Set up monitoring and management tools provided by your cloud platform to keep an eye on your Raspberry Pi's health and performance. This might include dashboards, alerts, and logging.
10. **Scale and Optimize:** As your project grows, you may need to scale your cloud resources and optimize your code and infrastructure for performance and cost-efficiency.